In [6]:
```python
# ============================================================
# Aircraft Metadata
# Enhanced Exploratory Data Analysis
# ============================================================

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings("ignore")

plt.style.use("ggplot")

pd.set_option("display.max_columns", None)

print("Libraries Imported Successfully")
```

```
Libraries Imported Successfully
```

In [7]:
```python
aircraft = pd.read_csv(
    "../../data/processed/aircraft/aircraft_metadata_final.csv",
    parse_dates=[
        "LAST ACTION DATE",
        "CERT ISSUE DATE",
        "AIR WORTH DATE",
        "EXPIRATION DATE"
    ]
)

print("Dataset Loaded Successfully")
```

```
Dataset Loaded Successfully
```

In [8]:
```python
print("="*60)
print("Aircraft Metadata Overview")
print("="*60)

print(f"Rows    : {aircraft.shape[0]:,}")
print(f"Columns : {aircraft.shape[1]}")
```

```
============================================================
Aircraft Metadata Overview
============================================================
Rows    : 314,417
Columns : 29
```

In [9]:
```python
aircraft.head()
```

Out[9]:

| | N-NUMBER | SERIAL NUMBER | MFR MDL CODE | ENG MFR MDL | YEAR MFR | TYPE REGISTRANT | LAST ACTION DATE | CERT ISSUE DATE | CERTIFIC |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 100 | 5334 | 7100510 | 17003.0 | 1940.0 | 1.0 | 2023-01-22 | 2005-05-06 | |
| **1** | 10000 | 10000 | 2130004 | NaN | NaN | 7.0 | 2024-08-23 | 2024-08-23 | |
| **2** | 10001 | A28 | 9601202 | 67007.0 | 1928.0 | 1.0 | 2023-07-18 | 2019-02-27 | |
| **3** | 10004 | T18208245 | 2072738 | NaN | NaN | 7.0 | 2023-07-22 | 2013-03-12 | |
| **4** | 10006 | BG-72 | 1152020 | 17026.0 | 1955.0 | 1.0 | 2023-04-21 | 1998-08-26 | |

In [10]:

```
top_manufacturers = (
    aircraft["MANUFACTURER"]
    .value_counts()
    .head(20)
)

top_manufacturers
```

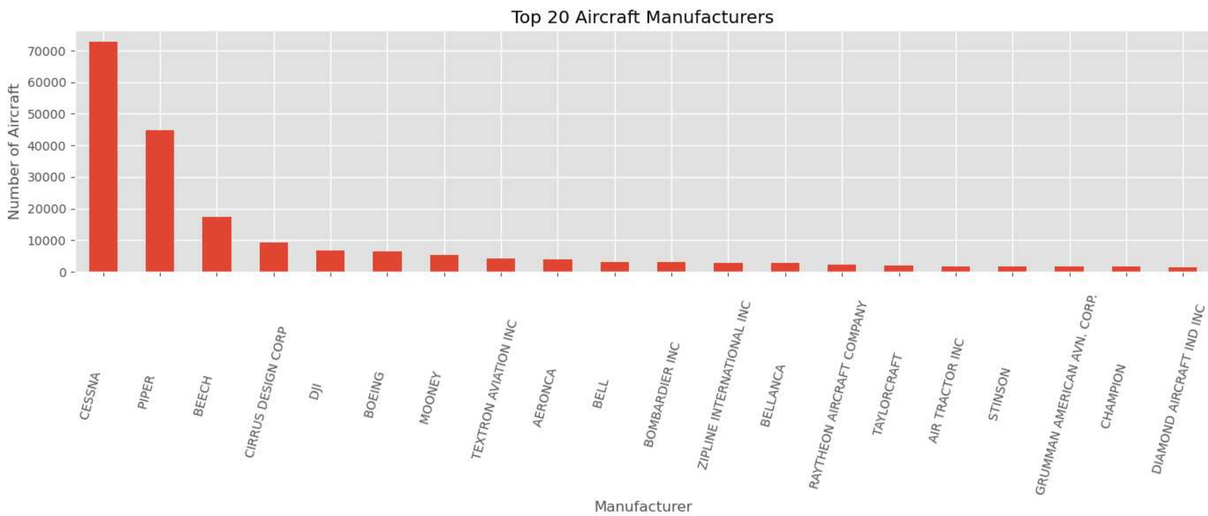
Out[10]:

```
MANUFACTURER
CESSNA                         72727
PIPER                          44842
BEECH                          17454
CIRRUS DESIGN CORP              9330
DJI                             6643
BOEING                          6549
MOONEY                          5392
TEXTRON AVIATION INC            4285
AERONCA                         3945
BELL                            3084
BOMBARDIER INC                  3000
ZIPLINE INTERNATIONAL INC       2730
BELLANCA                        2697
RAYTHEON AIRCRAFT COMPANY       2156
TAYLORCRAFT                     1897
AIR TRACTOR INC                 1798
STINSON                         1727
GRUMMAN AMERICAN AVN. CORP.     1674
CHAMPION                        1602
DIAMOND AIRCRAFT IND INC        1533
Name: count, dtype: int64
```

In [11]:

```
plt.figure(figsize=(14,6))

top_manufacturers.plot(kind="bar")
```

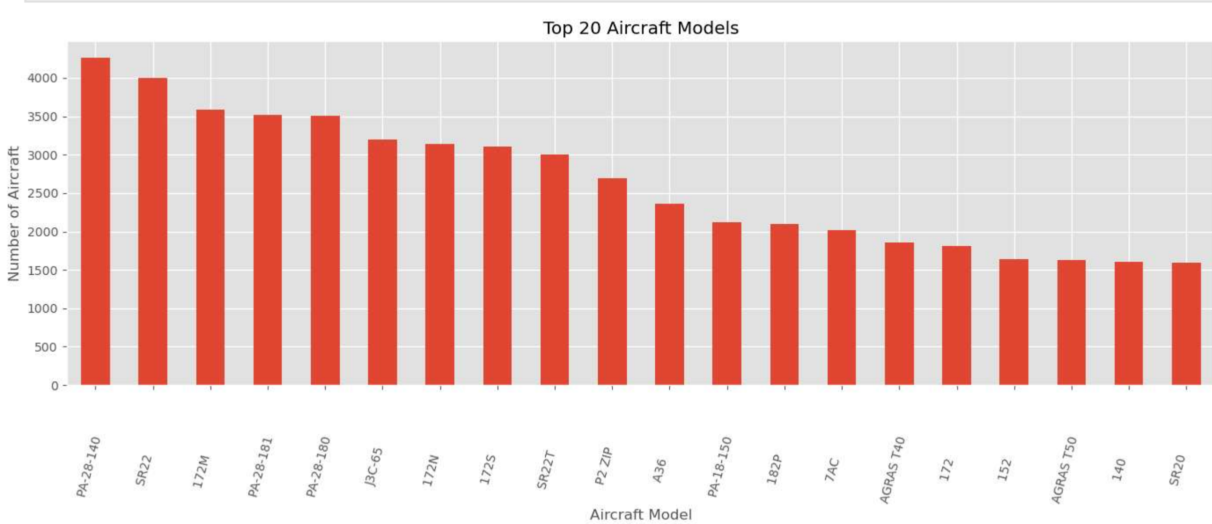
```
plt.title("Top 20 Aircraft Manufacturers")

plt.xlabel("Manufacturer")

plt.ylabel("Number of Aircraft")

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()
```




In [12]:
```
top_models = (
    aircraft["AIRCRAFT_MODEL"]
    .value_counts()
    .head(20)
)

top_models
```

Out[12]:
```
AIRCRAFT_MODEL
PA-28-140       4264
SR22            3997
172M            3582
PA-28-181       3523
PA-28-180       3508
J3C-65          3192
172N            3140
172S            3101
SR22T           3000
P2 ZIP          2689
A36             2359
PA-18-150       2119
182P            2100
7AC             2022
AGRAS T40       1854
172             1806
152             1636
AGRAS T50       1624
140             1603
SR20            1590
Name: count, dtype: int64
```

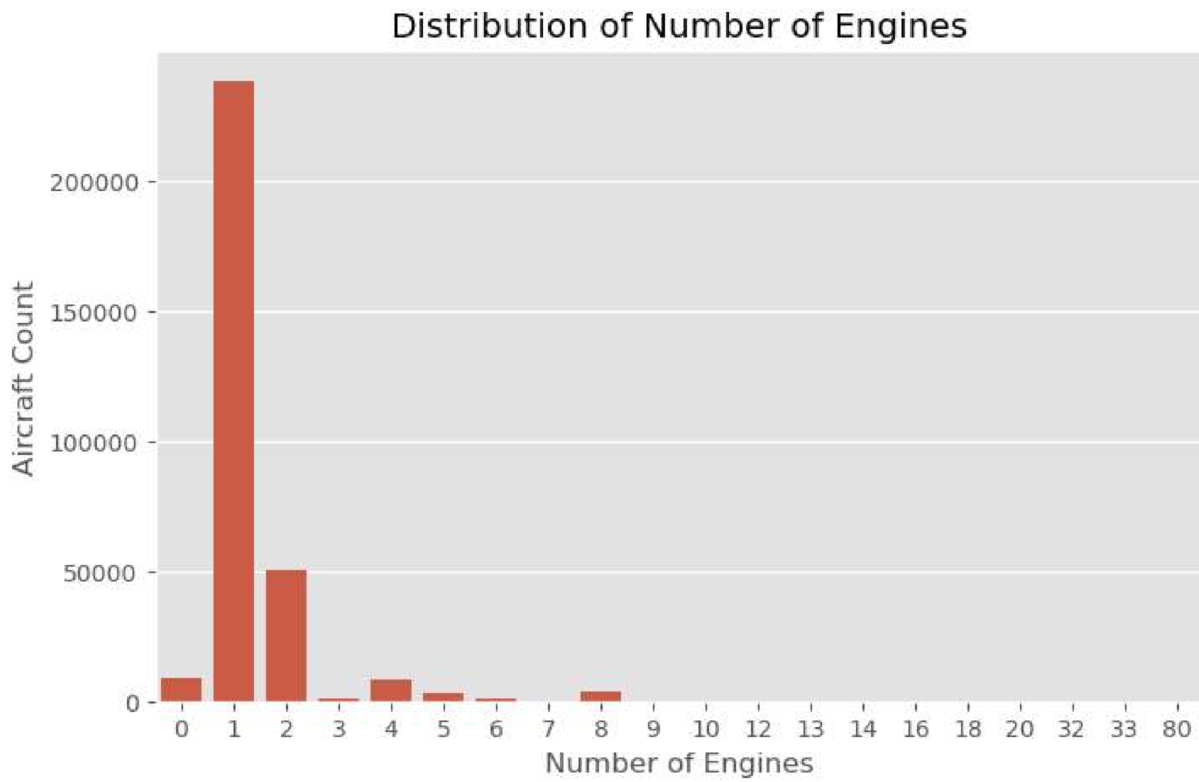
In [13]:
```python
plt.figure(figsize=(14,6))

top_models.plot(kind="bar")

plt.title("Top 20 Aircraft Models")

plt.xlabel("Aircraft Model")

plt.ylabel("Number of Aircraft")

plt.xticks(rotation=75)

plt.tight_layout()

plt.show()
```



In [14]:
```
aircraft["NUMBER_OF_ENGINES"].value_counts().sort_index()
```

Out[14]:
```
NUMBER_OF_ENGINES
0       9134
1     238092
2      50219
3       1094
4       7953
5       2843
6       1390
7          5
8       3606
9         22
10        15
12        18
13         3
14         1
16        12
18         1
20         5
32         1
33         1
80         2
Name: count, dtype: int64
```

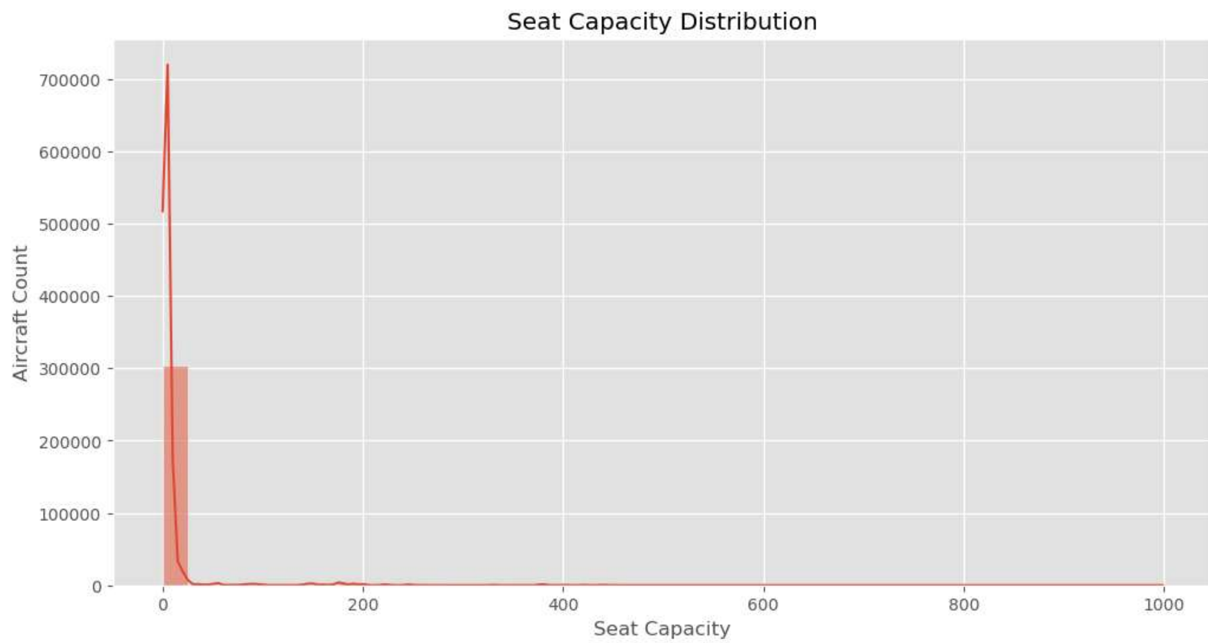
In [15]:
```
plt.figure(figsize=(8,5))

sns.countplot(
    data=aircraft,
    x="NUMBER_OF_ENGINES"
)

plt.title("Distribution of Number of Engines")

plt.xlabel("Number of Engines")

plt.ylabel("Aircraft Count")

plt.show()
```




In [16]:
```
aircraft["SEAT_CAPACITY"].describe()
```

Out[16]:
```
count    314417.000000
mean          9.686572
std          36.618971
min           0.000000
25%           2.000000
50%           4.000000
75%           5.000000
max         999.000000
Name: SEAT_CAPACITY, dtype: float64
```

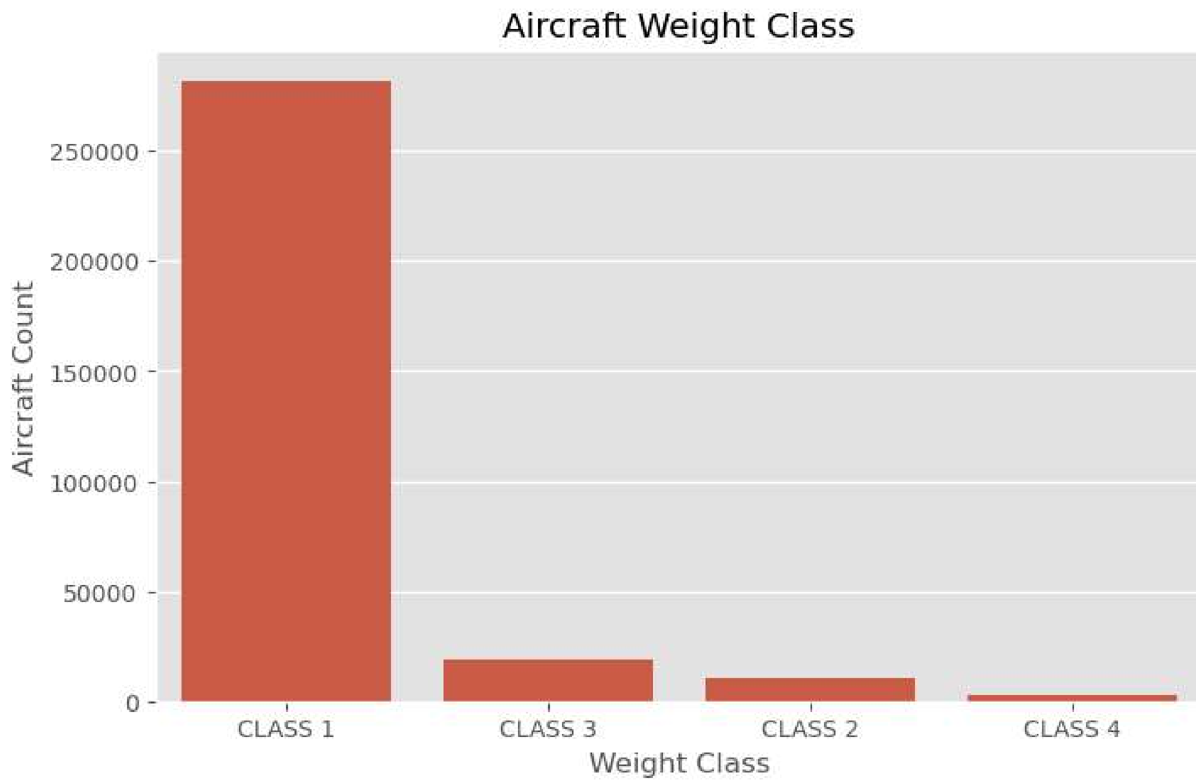
In [17]:
```
plt.figure(figsize=(12,6))

sns.histplot(
    aircraft["SEAT_CAPACITY"],
    bins=40,
    kde=True
)

plt.title("Seat Capacity Distribution")

plt.xlabel("Seat Capacity")

plt.ylabel("Aircraft Count")

plt.show()
```




In [18]:
```
aircraft["WEIGHT_CLASS"].value_counts()
```

Out[18]:
```
WEIGHT_CLASS
CLASS 1    281667
CLASS 3     19188
CLASS 2     10406
CLASS 4      3156
Name: count, dtype: int64
```

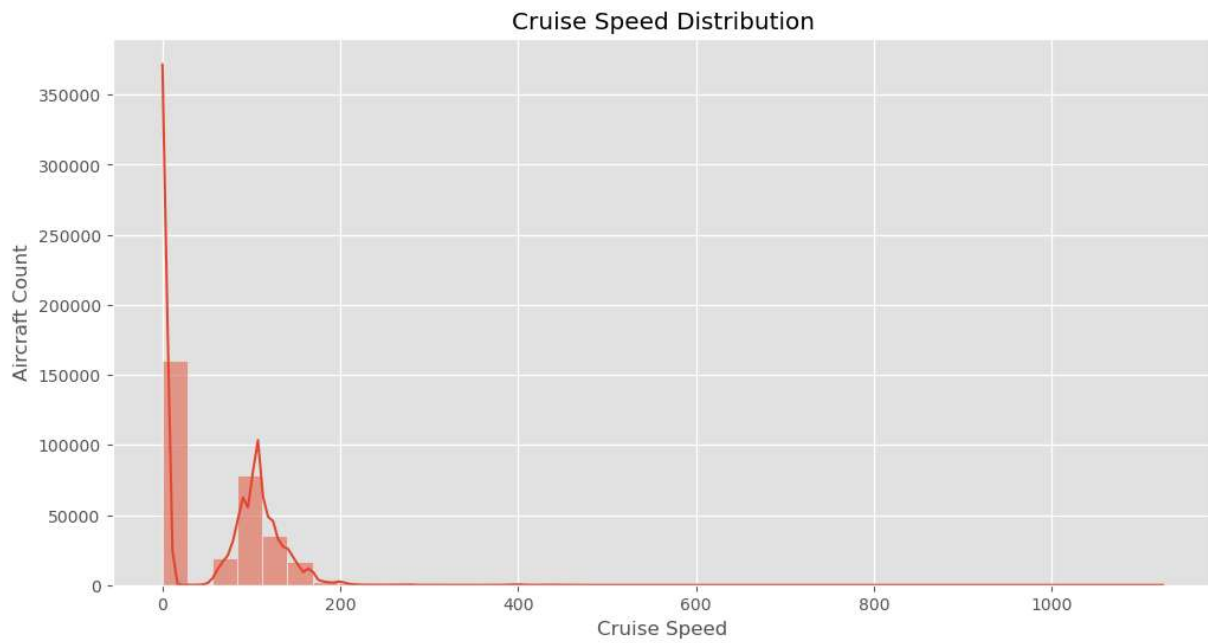
In [19]:
```
plt.figure(figsize=(8,5))

sns.countplot(
    data=aircraft,
    x="WEIGHT_CLASS",
    order=aircraft["WEIGHT_CLASS"].value_counts().index
)

plt.title("Aircraft Weight Class")

plt.xlabel("Weight Class")

plt.ylabel("Aircraft Count")

plt.show()
```




In [20]:
```
aircraft["CRUISE_SPEED"].describe()
```

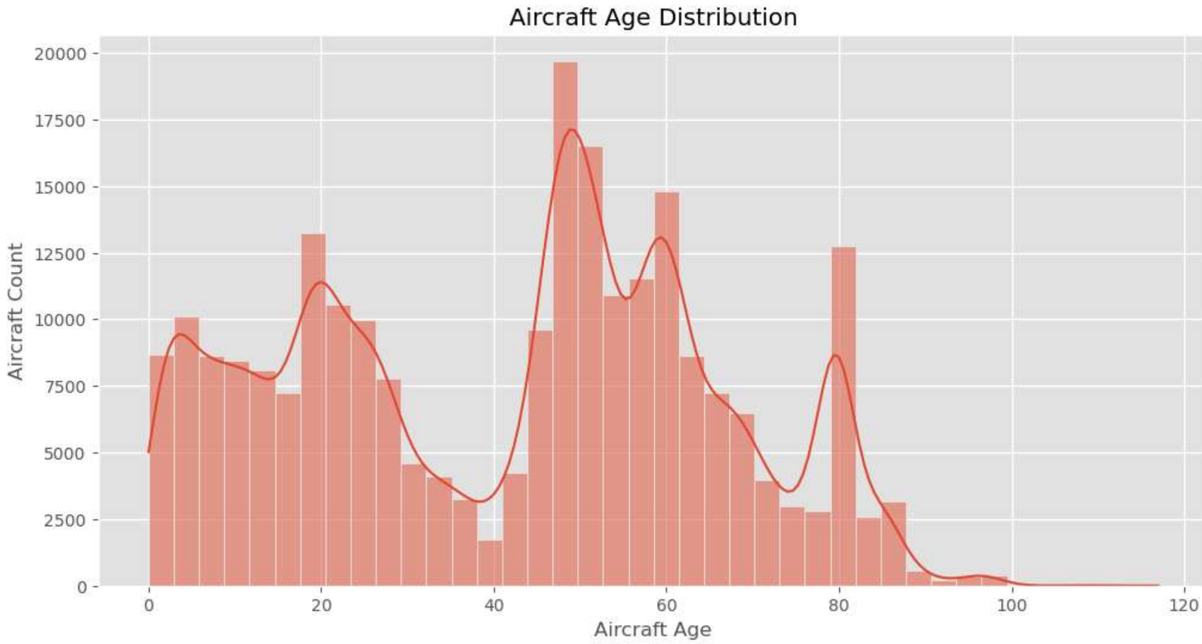
Out[20]:
```
count    314417.000000
mean         54.912139
std          60.885767
min           0.000000
25%           0.000000
50%           0.000000
75%         107.000000
max        1125.000000
Name: CRUISE_SPEED, dtype: float64
```

In [21]:
```
plt.figure(figsize=(12,6))

sns.histplot(
    aircraft["CRUISE_SPEED"],
    bins=40,
    kde=True
)

plt.title("Cruise Speed Distribution")

plt.xlabel("Cruise Speed")

plt.ylabel("Aircraft Count")

plt.show()
```

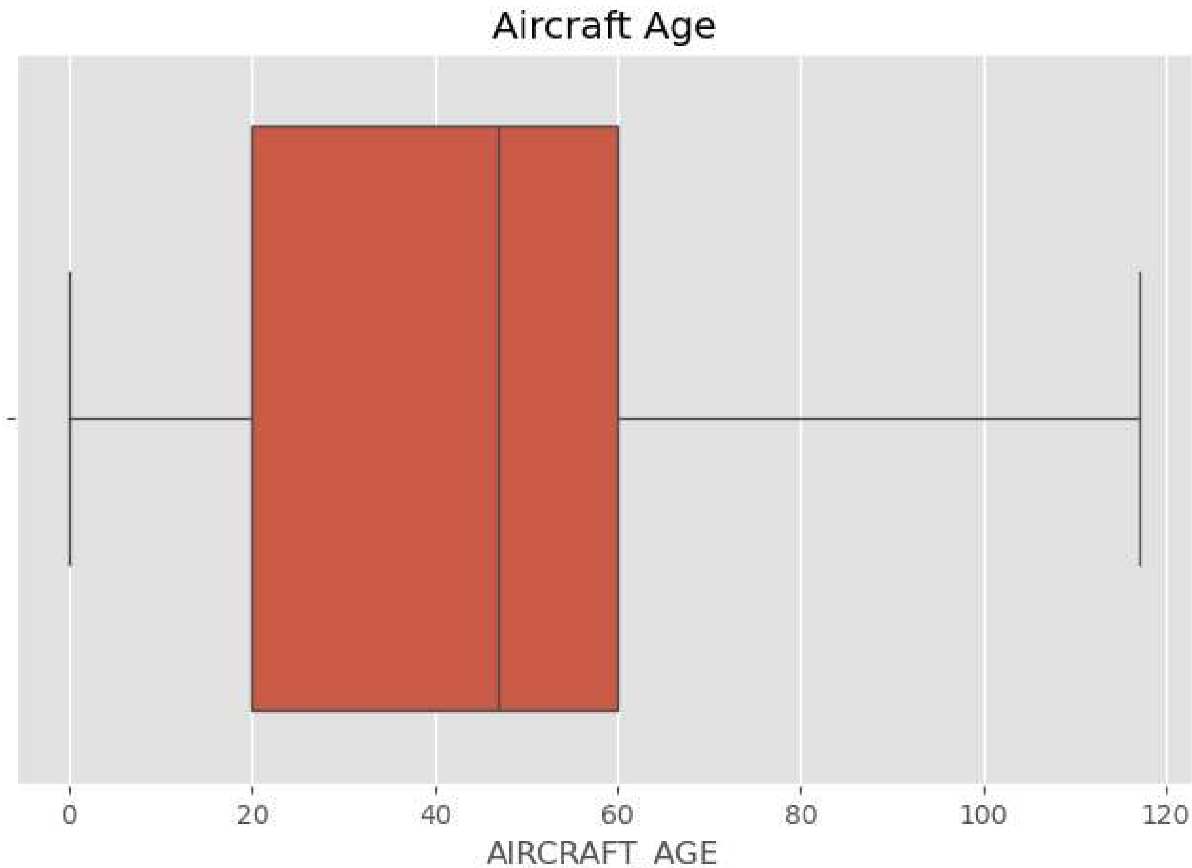



In [22]:
```
current_year = pd.Timestamp.now().year

aircraft["AIRCRAFT_AGE"] = (
    current_year -
    aircraft["YEAR MFR"]
)
```

In [23]:
```
plt.figure(figsize=(12,6))

sns.histplot(
    aircraft["AIRCRAFT_AGE"],
    bins=40,
    kde=True
)

plt.title("Aircraft Age Distribution")

plt.xlabel("Aircraft Age")

plt.ylabel("Aircraft Count")

plt.show()
```




In [24]:
```
plt.figure(figsize=(8,5))

sns.boxplot(
    x=aircraft["AIRCRAFT_AGE"]
)

plt.title("Aircraft Age")

plt.show()
```



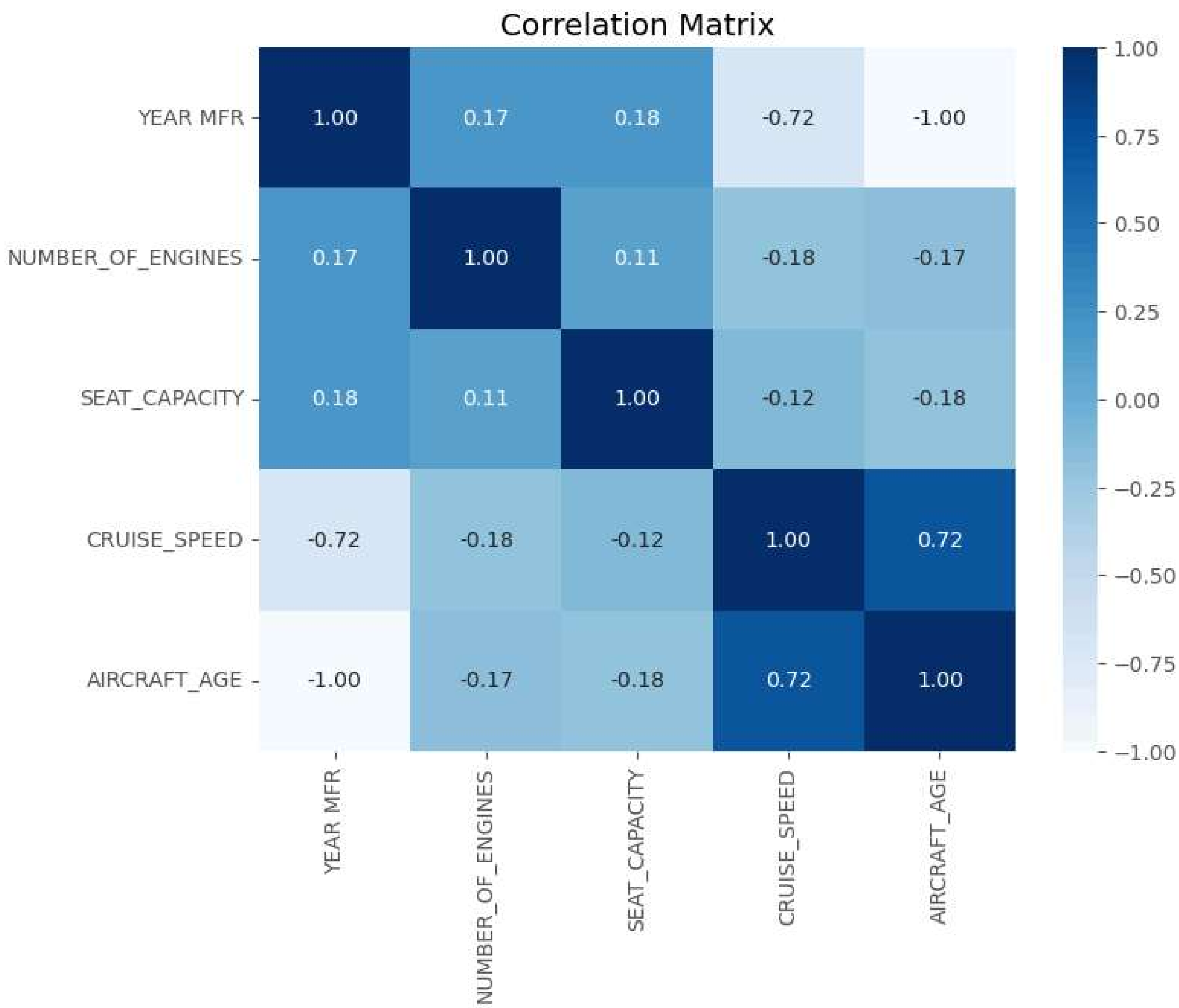

In [25]:
```
numeric = aircraft[[
    "YEAR MFR",
    "NUMBER_OF_ENGINES",
    "SEAT_CAPACITY",
    "CRUISE_SPEED",
    "AIRCRAFT_AGE"
]]

numeric.corr()
```

Out[25]:

| | YEAR MFR | NUMBER_OF_ENGINES | SEAT_CAPACITY | CRUISE_SPEED | A |
|---|---|---|---|---|---|
| **YEAR MFR** | 1.000000 | 0.172876 | 0.183268 | -0.717565 | |
| **NUMBER_OF_ENGINES** | 0.172876 | 1.000000 | 0.105280 | -0.181169 | |
| **SEAT_CAPACITY** | 0.183268 | 0.105280 | 1.000000 | -0.124968 | |
| **CRUISE_SPEED** | -0.717565 | -0.181169 | -0.124968 | 1.000000 | |
| **AIRCRAFT_AGE** | -1.000000 | -0.172876 | -0.183268 | 0.717565 | |

In [26]:
```
plt.figure(figsize=(8,6))

sns.heatmap(
    numeric.corr(),
```

```
    annot=True,
    cmap="Blues",
    fmt=".2f"
)

plt.title("Correlation Matrix")

plt.show()
```




In [27]:
```
missing = (
    aircraft.isnull().sum()
    .sort_values(ascending=False)
)

missing
```

Out[27]:
```
YEAR MFR                68411
AIRCRAFT_AGE            68411
AIR WORTH DATE          41979
ENG MFR MDL             37579
CERTIFICATION           33903
EXPIRATION DATE          7923
CERT ISSUE DATE          7891
TYPE REGISTRANT          1192
MFR MDL CODE                0
SERIAL NUMBER               0
TYPE AIRCRAFT               0
TYPE ENGINE                 0
N-NUMBER                    0
LAST ACTION DATE            0
MODE S CODE                 0
STATUS CODE                 0
MODE S CODE HEX             0
UNIQUE ID                   0
AIRCRAFT_MODEL              0
AIRCRAFT_TYPE_CODE          0
ENGINE_TYPE_CODE            0
MANUFACTURER                0
AIRCRAFT_CATEGORY           0
BUILD_CERTIFICATE           0
SEAT_CAPACITY               0
NUMBER_OF_ENGINES           0
WEIGHT_CLASS                0
CRUISE_SPEED                0
TYPE_CERTIFICATE            0
CERTIFICATE_HOLDER          0
dtype: int64
```

In [28]:
```
plt.figure(figsize=(10,6))

missing[missing>0].plot(kind="bar")

plt.title("Remaining Missing Values")

plt.ylabel("Count")

plt.show()
```

Remaining Missing Values
Count
70000
60000
50000
40000
30000
20000
10000
0
YEAR MFR
AIRCRAFT_AGE
AIR WORTH DATE
ENG MFR MDL
CERTIFICATION
EXPIRATION DATE
CERT ISSUE DATE
TYPE REGISTRANT